PowerPoint Presentation: Chapter 9 - Prevention of Pneumonia: Vaccination Strategies

Dr. Siddalingaiah H S - MBBS Phase 3 Part 1 Integration Class

Slide 1: Title Slide

Pneumonia Treatment & Immunization CBME Module Chapter 9: Prevention of Pneumonia - Vaccination Strategies

Dr. Siddalingaiah H S Associate Professor, Community Medicine MBBS Phase 3 Part 1 Integration Class - Day 5

Slide 2: Learning Objectives

By end of this session, MBBS students will be able to:

1. Analyze global and Indian pneumonia burden and preventable deaths
2. Understand pneumococcal and influenza vaccination strategies
3. Apply age-appropriate vaccination recommendations
4. Demonstrate effective vaccine counseling techniques
5. Implement vaccination programs in clinical practice

CBME Competencies: Medical Knowledge, Patient Care, Systems-Based Practice

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Slide 3: Pneumonia Prevention Paradigm Shift

Vaccination is the most effective pneumonia prevention strategy

- **Before 2000:** Focus on treatment and antibiotics
- **After PCV introduction:** Dramatic reduction in invasive disease
- **Current era:** Comprehensive immunization + improved care
- **Future vision:** Pneumonia elimination through vaccination

Timeline graphic showing paradigm shift

Slide 4: Global Pneumonia Burden - Vaccine-Preventable

📊 **Annual Impact:**

- **2.5 million deaths worldwide** (leading infectious cause)
- **1.5 million deaths preventable** through vaccination
- **50-80% reduction possible** with optimal vaccination
- **Economic savings:** $27-61 returned per dollar invested

Global map with country-wise pneumonia burden

Slide 5: Indian Context - Pneumonia Mortality

IN **Indian Reality:**

- **340,000 deaths annually** from pneumonia
- **25-30% of deaths vaccine-preventable**
- **17% of under-5 deaths** attributed to pneumonia
- **Highest burden:** Uttar Pradesh, Bihar, Rajasthan

Indian pneumonia burden statistics with state-level data

Slide 6: Pneumococcal Disease Epidemiology

Streptococcus pneumoniae - The Primary Target

- **Most common bacterial pneumonia** cause worldwide
- **90+ serotypes** with varying pathogenicity
- **Invasive disease**: Pneumonia, meningitis, bacteremia
- **Nasopharyngeal carriage**: 10-40% in healthy adults

Microorganism image with serotype distribution

Slide 7: Pneumococcal Vaccine Impact Evidence

🎯 **Proven Effectiveness:**

- **Direct protection:** 75-90% reduction in invasive disease
- **Herd immunity:** Protection of unvaccinated populations
- **Serotype replacement:** Emergence of non-vaccine strains
- **Cost-effective:** <$100 per DALY averted

Before-after graphs showing vaccine impact

Slide 8: Influenza as Secondary Pneumonia

Seasonal Influenza Contribution:

- **300-500,000 deaths annually** worldwide
- **Bacterial superinfection:** 10-15% lead to pneumonia
- **High-risk groups:** Elderly, immunocompromised, pregnant
- **Secondary bacterial pneumonia:** S. aureus, S. pneumoniae

Venn diagram showing viral-bacterial interaction

Slide 9: Comprehensive Vaccination Strategy

🎯 **Three Pillars of Pneumonia Prevention:**

1. **Pneumococcal Vaccination** - Primary bacterial prevention
2. **Influenza Vaccination** - Viral pneumonia prevention
3. **Adjunct Measures** - Hygiene, nutrition, air quality

Three-pillars model diagram

Slide 10: Pneumococcal Vaccine Types - Overview

Vaccine	Age Groups	Serotypes	Mechanism	Schedule
PCV13	<6 months + adults	13 (including 19A)	Conjugate (T-cell)	2,4,6 +12-15m
PPSV23	≥2 years + elderly	23 additional	Polysaccharide	Single dose

Differentiated vaccine comparison table

Slide 11: PCV13 Schedule and Administration

Pneumococcal Conjugate Vaccine (Prevnar 13)

Pediatric Schedule:

- **Primary series:** 2, 4, 6 months
- **Booster:** 12-15 months
- **Catch-up:** Up to 59 months for high-risk

Administration:

- **Route:** IM deltoid/thigh
- **Dose:** 0.5 mL
- **Storage:** 2-8°C
- **Contraindications:** Severe allergy

Visual schedule timeline

Slide 12: PPSV23 Recommendations

Pneumococcal Polysaccharide Vaccine (Pneumovax 23)

Target Groups:

- **Age ≥65 years:** Routine recommendation
- **High-risk adults:** Chronic diseases, immunocompromised
- **Special groups:** Asplenic patients, transplant recipients

Administration:

- **Route:** IM/SC deltoid
- **Timing:** 8+ weeks after PCV13
- **Re-vaccination:** Every 5 years in high-risk

- **Precautions:** Immune response may be reduced

Target groups illustration

Slide 13: Sequential Adult Vaccination

Optimal Adult Strategy: PCV13 → PPSV23

Why Sequential Approach:

- **Broader coverage:** All vaccine serotypes protected
- **Immune response:** Conjugate vaccine enhances polysaccharide response
- **Duration of protection:** 10+ years for PCV13, 5 years for PPSV23
- **Cost-effectiveness:** Systematic approach maximizes benefit

Sequential vaccination flowchart

Slide 14: Influenza Vaccine Seasons & Strains

Annual Vaccination - Evolving Strains

Northern Hemisphere:

- **Season:** December-March peaks
- **2025-2026 predict:** H1N1, H3N2, 2 B lineages

Southern Hemisphere:

- **Season:** April-September
- **Strain prediction:** Basis for Northern vaccines

Quadrant showing seasonal variations

Slide 15: Influenza Vaccine Types Comparison

Type	Strains Covered	Age Groups	Advantages	Limitations
Quadrivalent	4 strains (2A + 2B)	≥6 months	Best coverage	Higher cost
High-dose	4× antigen (elderly)	≥65 years	Better response	Local reactions
Recombinant	Cell culture	≥18 years	No egg allergy	Newer technology

Comparison matrix with visuals

Slide 16: Influenza Vaccination Timing & Coverage

Optimal Timing:

- **Start:** September (Northern hemisphere)

- **Peak season:** December-March
- **Minimum protection:** 2 weeks after vaccination
- **Annual requirement:** Strains change yearly

High-Risk Groups (Priority):

1. **Pregnant women** (all trimesters)
2. **Children:** 6-59 months
3. **Elderly:** ≥65 years
4. **Chronically ill:** Cardiac, pulmonary, renal
5. **Healthcare workers:** Patient protection

Priority pyramid visualization

Slide 17: Vaccine Effectiveness Data

Pneumococcal Vaccine VE:

- **Invasive disease:** 75-90% reduction
- **Pneumonia hospitalization:** 40-60% in elderly
- **All-cause mortality:** 30-50% reduction
- **Herd protection:** 35-50% in unvaccinated

Influenza Vaccine VE:

- **Healthy adults:** 40-60% against illness
- **Elderly:** 50-70% against severe outcomes
- **Children:** 60-75% effectiveness demonstrated

Effectiveness bar graphs with confidence intervals

Slide 18: Cost-Effectiveness Analysis

Economic Justification:

- **PCV introduction:** $27-61 return per dollar invested
- **Global impact:** $400 billion savings annually
- **Break-even price:** $10-30 per vaccine dose
- **Cost per DALY:** <$100 (highly cost-effective)

Cost-benefit analysis graph

Slide 19: Vaccine Counseling Framework

EMPATHY Communication Model

Explore concerns actively **M**ake scientific information clear **P**ersonalize risk-benefit discussion **A**ppear credible and confident **T**ailor information to literacy level/culture **H**arness social networks and influencers **I**ncrease access and convenience **Z**ero regrets approach to decision-making

8-step model with icons

Slide 20: Addressing Vaccine Hesitancy

Common Concerns & Responses:

"Vaccines cause disease"

- *Response:* Inactivated vaccines cannot cause disease

"Natural immunity is better"

- *Response:* Natural infection carries higher risks

"Too many vaccines overwhelm"

- *Response:* Immune system handles billions of antigens naturally

"Autism link"

- *Response:* Multiple studies show no causal relationship

Myth-busting chart with evidence-based responses

Slide 21: Implementation Strategies

Healthcare Settings:

- **Primary care:** Routine vaccination opportunities
- **Hospital admission:** Missed vaccination catch-up
- **Emergency department:** Vaccine counseling opportunities
- **Outpatient clinics:** Designated immunization days

Community Approaches:

- **School-based programs:** High coverage in educational settings
- **Mobile teams:** Hard-to-reach areas coverage
- **Campaigns:** National immunization days
- **Reminder systems:** SMS, app-based notifications

Integration model diagram

Slide 22: Program Challenges & Solutions

Coverage Barriers:

- **Access issues:** Transport, cost, availability
- **Awareness gaps:** Lack of knowledge, myths
- **Health system:** Fragmented delivery, stock-outs
- **Policy gaps:** Insufficient funding, prioritization

Solutions:

- **Mobile outreach:** Community-based service delivery
- **Communication campaigns:** Targeted education efforts
- **Cold chain strengthening:** Reliable vaccine supply
- **Digital integration:** CoWIN platform optimization

Challenge-solution matrix

Slide 23: Monitoring & Evaluation

Key Performance Indicators:

- **Coverage rates:** ≥90% across all indicators
- **Drop-out rates:** <10% between vaccine doses
- **Timely vaccination:** ≥85% given on schedule
- **AEFI reporting:** 100% of serious events investigated

Dashboard-style KPI visualization

Slide 24: CBME Assessment Questions

Question: Most cost-effective pneumonia prevention strategy?

- A) Hospital treatment protocols
- B) **Vaccination programs** ★
- C) Antibiotic stewardship
- D) Air pollution control

Question: Which pneumococcal vaccine provides longest immunity?

- A) PCV7 (discontinued)
- B) **PCV13** ★
- C) PPSV23
- D) None of the above

Slide 25: Clinical Case Application

Patient Case: 65-year-old diabetic

- **Presentation:** Productive cough, fever, shortness of breath
- **History:** Not vaccinated against pneumococcus or influenza
- **Treatment:** Appropriate antibiotics + vaccination counseling
- **Prevention:** PCV13 now + influenza vaccination timing

Learning Point: Every pneumonia admission is vaccination opportunity

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Slide 26: Future Directions

Advanced Vaccine Development:

- **Universal pneumococcal:** Conserved antigens approach
- **Maternal immunization:** Prenatal antibody transfer
- **LAIV improvements:** Better efficacy in all age groups
- **Combination vaccines:** Single-shot pneumonia prevention

Digital Integration:

- **Blockchain certificates:** Tamper-proof vaccination records
- **AI prediction:** Outbreak forecasting and vaccine stock optimization
- **Mobile apps:** Real-time coverage monitoring and reminders

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Slide 27: Key Takeaways

🧠 **Essential Concepts:**

1. **Vaccination prevents 50-80% of pneumonia deaths** globally
2. **PCV13 + PPSV23 sequential strategy** for optimal adult protection
3. **Annual influenza vaccination** essential for high-risk groups
4. **Cost-benefit ratio exceeds 50:1** for vaccination programs
5. **Herd immunity protects vulnerable** unvaccinated populations
6. **Vaccine counseling requires empathy** and evidence-based responses
7. **Implementation barriers** can be overcome with innovative solutions

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Slide 28: Session Evaluation

Rate your learning experience:

- **Knowledge acquired:** ★★★★★
- **Clinical relevance:** ★★★★★
- **Teaching quality:** ★★★★★
- **Materials usefulness:** ★★★★★

What was most valuable today? What needs more emphasis? Additional topics to cover?

Slide 29: References & Resources

📚 **Key Guidelines:**

1. WHO Pneumococcal Vaccine Position Paper (2022)
2. IAP Pediatric Consensus Guidelines (2021)
3. ACIP Adult Immunization Schedule (2024)
4. CDC Pink Book: Epidemiology and Prevention of Vaccine-Preventable Diseases

🔗 **Online Resources:**

- WHO Immunization: https://www.who.int/immunization

- CDC Vaccines: https://www.cdc.gov/vaccines
- IAP Guidelines: https://iapindia.org/guidelines

Slide 30: Next Session Preview

Chapter 10: Pneumococcal Vaccination - Detailed Types & Recommendations

Key Topics:

- PCV13 vs PPSV23 mechanisms
- Special populations schedules
- Safety and contraindications
- Implementation challenges

Next Class: October 15, 2025 | 9:00 AM **Location:** Integration Class Hall

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Slide 31: Q&A and Discussion

Questions & Interactive Discussion

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Interactive Q&A with audience participation

Presentation Developed for: Pneumonia Treatment & Immunization CBME Module MBBS Phase 3 Part 1 Integration Class

Total Slides: 31 Duration: 50-60 minutes Faculty: Dr. H S Lingaiah, FIAHS Target: Medical students integrating preventive medicine